Centre for Brain, Mind and Markets

THE UNIVERSITY OF MELBOURNE | FACULTY OF BUSINESS & ECONOMICS

FNCE30010
Algorithmic Trading

Semester 2, 2026
Project 2 (Backtesting)
Report due on **11:59 pm on Sunday, 25 October 2025**

Nitin Yadav
Centre for Brain, Mind & Markets Research.
Department of Finance.
Faculty of Business and Economics.
The University of Melbourne.

Administrative Arrangements

Report due date:	**11:59 pm on Sunday, 25 October 2026**
Late assignments:	Late submission assignments may attract a penalty unless an extension has been granted. All extension requests must be made prior to the assessment due date and supported with appropriate documentation. Unless an extension has been granted, penalties to the assessment will be applied. For assignments submitted after the due date, the mark awarded will be reduced by 10% for each day the work is late. For assignments submitted later than 5 working days after the due date will not be marked and will receive no marks.
Where to submit:	Assignment submission is via the LMS Assignment Submission link for all written assignments. Please refer to the LMS Student Guide: Turnitin Assignments for detailed submission instructions, if needed.
Group size:	For undergraduates the maximum number in each group is 2. For postgraduates this is an individual assessment.
Submission Metadata:	You should include your details as part of your submission. **Subject Code and Name** **Student Names and Numbers of all members in group** **Assignment Name**
Word limit:	Maximum of 1800 words (excluding Appendix).
Marks:	This assignment counts 25%(UG)/30%(PG) towards the final mark in this subject.
Gen AI	***Use of Gen AI tools is allowed. Please cite them appropriately.***

The assignment must be your group's own work. Students are encouraged to discuss the assignment and to share information sources. However, the writing of each group's assignment must be conducted separately and independently.

Assignment Instructions

Background: Background: You are hired on probation as an algorithmic trader in one of the top "Quant" firms. Your first task is to explore the trading strategies available in the public domain with the objective to improvise on it. You should select a strategy, analyse it, understand its financial underpinnings, and then report on your findings. You are also free to use LLM tools such as ChatGPT, Claude, Gemini, etc to come up with ideas to test.

Note that your manager has more than 25 years of experience in the trading/investment industry and she is only convinced by sound scientific arguments and evidence.

Your main focus should be motivating the strategy and suggested improvements from a finance/economics underpinnings. You are free to use publicly available resources such as strategy explorer from QuantConnect, or any strategy published in a blog/academic journal.

Assignment tasks:

(1) Select and describe the selected trading strategy (e.g., technical, statistical, fundamental, etc).

(2) Discuss potential weaknesses in the selected strategy, suggestions to improve on it, and back test it. Note improvisation does not mean that you change the value of the parameters (e.g., changing the length of the short moving average from 5 to 3).

(3) Discuss the improvised strategy along with experimental results.

Marks will be allocated as follows:

- Strategy description and improvisation: 10 marks
- Methodology and evaluation: 15 marks
- Writing style: 5 marks